The Erdős–Graham two-set problem, II: the general case

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Abstract

Call a set $S \subseteq \mathbb{Z}^+$ *3-permutable* if it admits an arrangement (a bijection $\mathbb{N} \to S$) containing no monotone 3-term arithmetic progression. Davis, Entringer, Graham and Simmons (1977) partitioned $\mathbb{Z}^+$ into three 3-permutable sets and asked whether two suffice (Erdős–Graham 1979; problem #197). Part I of this series proved that the canonical dyadic partition fails. This paper studies the general problem. Unconditionally, we prove a *hereditary splitness* theorem: if (A, B) is a partition of $\mathbb{Z}^+$ with both parts 3-permutable, then for every $k \geq 1$ and every residue c mod 2^k, the class-c section of all but finitely many dyadic blocks meets both parts. In particular no 3-permutable set contains infinitely many full dyadic blocks, none contains a full mod-4 class section of infinitely many blocks, none contains an infinite arithmetic progression, and every blockwise "lattice" partition — with one part's block minority a full residue class modulo a power of two, under *any* law of which part owns the class at each scale — fails. The proof runs through a new uniform-in-p rigidity theorem for a three-axiom diagonal core $C3(p)$ on $(M, 2M]$, an affine restriction transport, and a density argument for diagonal pairs. Conditionally, we assemble a proof that *no* two-set partition works, modulo three explicitly stated open statements whose exact machine-verified status we document: a punctured-rung growth law, a coupled two-seam core closure, and a supply-cap (affordability) statement that we isolate as the problem's terminal open question.

Status of this document

This is an assembly skeleton (campaign notes/86 §3): section plan, full theorem statements, dependency graph, and per-section pointers to the campaign notes that already contain the finished mathematics. Status tags: **[PROVED]** = complete hand proof in the named note, audit-cleared where audited; **[MACHINE-CHECKED]** = machine-verified at the stated scales with per-instance schema; **[GAP]** = open statement, stated here as a numbered Hypothesis. The three Hypotheses (13, 20, 22) are the *only* open inputs; every other statement in this skeleton is **[PROVED]**.

*With extensive machine assistance from Claude (Anthropic) and from the SAT solvers CaDiCaL and OR-Tools CP-SAT; SAT certificates, code, and a one-command reproduction kit are available in the accompanying repository (see Data Availability). All proofs have been verified by hand unless explicitly marked as machine-checked.

1 Introduction

Problem statement; history (DEGS 1977, Erdős–Graham 1979, Geneson 2018, LeSaulnier–Vijay 2010, Hirose–Saito); summary of Part I (the canonical dyadic partition fails: order-gadget reduction + $C3$ core rigidity at $p = 5$); the two main results of this paper:

- **Theorem A** (unconditional): the hereditary-splitness package — Lemma Q, Theorem ALT-DEAD, Corollary HSPLIT (§5), resting on Theorem $C3(p)$ (§3) and the Case-1 bridge at zero dust (§4).
- **Theorem B** (conditional assembly): no two-set partition of $\mathbb{Z}^+$ has both parts 3-permutable, modulo Hypotheses 13–22 (§9).

Source: notes/50 (final dependency graph); paper/main.tex §1 for the shared history; notes/86 §2 (definitive gap list). Prose to be written fresh.

2 Preliminaries and the block dichotomy

Arrangements, monotone 3-APs, 3-permutability; the midpoint-extremal rules R1–R4; dyadic blocks $B(t) = (2^t, 2^{t+1}]$; C-clean blocks; the rung theory $\mathrm{R}(a_1, a_2; M, D)$ (AP theory of $(M, 2M] \setminus D$ plus all guarded attack units $(t_{a-2j} \prec b_j)$ of the pair $\{a_1, a_2\}$, where $b_j := M + j$, $t_i := 2M - i$).

Proposition 1 (Block dichotomy, N4)**.** *For every partition (A, B) of $\mathbb{Z}^+$ exactly one holds: (Case 1) some part has infinitely many C_0-clean dyadic blocks for some constant C_0; (Case 2) both parts' in-block presence diverges.* ***[PROVED]***

Source: notes/43 §2, notes/46 §4A; transcription straightforward.

Supporting unconditional results imported from Part I: the orbit obstruction (lem:orbit), no-infinite-AP for finite reflector sets, the tower characterization, the $\times 4$ restriction lemma. **[PROVED, Part I]**

3 The affine core theorem $C3(p)$

The paper's engine. Order $\prec$ on $(M, 2M]$, AP-free = no monotone in-block 3-AP; $m_0 := 3M/2$; for odd $p \geq 5$ the *diagonal core* is

$$C3(p) \;=\; \{\, t_p \prec b_p,\;\; t_{p-2} \prec b_{p+1},\;\; t_{p+5} \prec b_{p-2} \,\}$$

(the attack units $j = p$, $p + 1$ of attacker $3p$ and $j = p - 2$ of attacker $3p + 1$).

Lemma 2 (Zigzag; phase dichotomy)**.** *Every d-ladder of a linear order is globally in one of two zigzag phases; adjacent rungs alternate leader/trailer.* ***[PROVED]***

Lemma 3 (Transfer lock $E(p)$)**.** *For even $M \geq 2p + 2$, on the odd $d = 2$ ladder the four odd core values sit in two adjacent index pairs, and: $M \equiv 0 \pmod 4$: $b_p \prec b_{p-2} \iff t_{p-2} \prec t_p$; $M \equiv 2 \pmod 4$: the lock reverses.* ***[PROVED]***

Lemma 4 (Flood P). *The mirror/zigzag induction at a center c over a g-class (POLAR, P2, G4 instances), phase-blind.* ***[PROVED]***

Theorem 5 ($L1(p)$). *Let $p \geq 5$ be odd, $M \equiv 0 \pmod 4$, $M \geq p+7$, and let $\prec$ be an AP-free order of $(M, 2M]$ satisfying $A2(p)$ and $A3(p)$. Then $b_p \prec b_{p-2}$ and $t_{p-2} \prec t_p$.* ***[PROVED]***

Theorem 6 (FLIP(p)). *Let $p \geq 5$ be odd, $M \equiv 2p+6 \pmod 8$, $M \geq 2p+6$, and let $\prec$ be AP-free satisfying $A2(p)$, $A3(p)$ and $b_p \prec b_{p-2}$. Then $A1(p)$ is impossible.* ***[PROVED]***

Theorem 7 ($C3(p)$ rigidity). *For every odd $p \geq 5$ and every $M \equiv 2p+6 \pmod 8$ with $M \geq 2p+6$, no AP-free order of $(M, 2M]$ satisfies $C3(p)$. Moreover the flip class is exact: on the complementary even class $C3(p)$ is satisfiable at every tested p* ***[MACHINE-CHECKED]****. For $p \equiv 1 \pmod 4$ the flip class is $M \equiv 0 \pmod 8$, which contains every dyadic scale $2^m \geq 2p+6$.* ***[PROVED]***

Source: notes/78 Part I (the affine write-up), toolkit notes/33 §§2–3; referee prose pass DONE, notes/86 §1 (one display fix: FLIP(p) Case I step 3, "$M > 2p+2$" → "$M > 2p-2$"). Machine record: e123 schema $p = 5..25$, e180 exact boundary scan, solver cross-validation at 9 odd p-values, identity layer re-derived to $p = 39$ (notes/80 §1.1). $p = 5$ instance = Part I thm:c3core.

4 Case 1: pigeonhole, density, and the bridge

Lemma 8 (PIN). *Let T be 3-permutable, $a_1 < a_2 \in T$ fixed, and $\mathbb{M}$ an infinite set of scales such that each $B(m)$, $m \in \mathbb{M}$, is C-clean for T and $2^m > a_2$. Then for all but finitely many $m \in \mathbb{M}$ the theory $\mathrm{R}(a_1, a_2; 2^m, D_m)$ is consistent (D_m the in-block dust).* ***[PROVED]***

Lemma 9 (DIAG-DENSE). *The number of diagonal usable pairs $\{3p, 3p+1\}$, $p \equiv 1 \pmod 4$, contained in $B(m)$ is at least $(2^m - 13)/12$.* ***[PROVED]***

Definition 10 (Rung hypothesis (H1)). For every $p \equiv 1 \pmod 4$, $p \geq 5$, and every $C \geq 0$ there is $m^*(p, C)$ such that for every $m \geq m^*(p, C)$ and every $D \subseteq B(m)$ with $|D| \leq C$, the theory $\mathrm{R}(3p, 3p+1; 2^m, D)$ is inconsistent.

Theorem 11 (B1). *Assume (H1). If $T \subseteq \mathbb{Z}^+$ has infinitely many C_0-clean dyadic blocks for some constant C_0, then T is not 3-permutable. No hypothesis on the partner is used.* ***[PROVED modulo (H1)]***

Corollary 12 (B1 at zero dust — unconditional). *(H1) at $C = 0$ is exactly Theorem 7 (the three $C3(p)$ units are among the rung's fired units). Hence, unconditionally: no 3-permutable set contains infinitely many full dyadic blocks.* ***[PROVED]***

The general-C case of (H1) is the first open input:

Hypothesis 13 (N3-GROW, the $\geq$-side (N3-b)). For every odd $x \geq 11$, all large M, and every puncture set D with $|D| < \lfloor (x-1)/4 \rfloor$, the theory $\mathrm{R}(x; M) \setminus D$ is inconsistent. **[GAP]**

Source: notes/52 (PIN §1.3, DIAG-DENSE §2.1, B1 §3.1) + Step-1 patch notes/74 §I.4; audits notes/60/60-1. Hypothesis 13: exact statement, proof skeleton LANE + SEV + (N3-b′), and the machine record (law $d^*(x) = \lfloor (x-1)/4 \rfloor$ exact and global at $x = 11/15/19/23/27$) in notes/74 Part I and notes/78 Part II. The $\leq$-side (N3-a) is [**PROVED**] modulo the single half-scale hypothesis GAP-SA-HALF (sharpness only, not consumed here). Supporting: Theorem N2-COMPLETE (every adjacent pair fires at all residues, $M \geq x + 57$), notes/73 [**MACHINE-CHECKED + partial schema**].

5 Hereditary 2-adic splitness (Theorem A)

The paper's unconditional headline. For $c \in \{0, 1, 2, 3\}$ let $\Lambda_c(t) := \{v \in B(t) : v \equiv c \pmod 4\}$.

Lemma 14 (Q, quarter-tail)**.** *No 3-permutable set contains $\Lambda_c(t)$ for one fixed c and infinitely many t.* ***[PROVED]***

Proof shape (one page): the 4-adic chart $\varphi(x) = (x - c + 4)/4$ (resp. $x/4$) maps $\Lambda_c(t)$ exactly onto $B(t-2)$; restriction preserves permutability, affine maps preserve 3-APs in both directions; the image has infinitely many full dyadic blocks — contradiction with Corollary 12.

Theorem 15 (ALT-DEAD)**.** *Call a scale t 4-pure for a partition (A, B) if some mod-4 class is monochromatic on $B(t)$. If infinitely many scales are 4-pure, then A and B are not both 3-permutable.* ***[PROVED]***

Corollary 16 (Lattice colorings die at every ownership law)**.** *Every blockwise mod-4 lattice coloring is dead — constant, alternating at any run-length law, unbounded-run, procrastinating ownership all included; every punctured near-lattice with on-class dust is dead at any puncture count.* ***[PROVED]***

Corollary 17 (HSPLIT, hereditary splitness)**.** *For every partition (A, B) with both parts 3-permutable, every $k \geq 1$ and every $c \bmod 2^k$, the section $(c + 2^k\mathbb{Z}) \cap B(t)$ is bichromatic for all but finitely many t. In particular every eventually 2^k-periodic block minority is dead.* ***[PROVED]***

Corollary 18 (L-NOTAIL)**.** *No 3-permutable set contains an infinite arithmetic progression.* ***[PROVED]*** *(two proofs: DEGS77 + restriction + affine invariance; or as a corollary of Lemma Q on the lattice family.)*

Remark 19 (Q-ODD, stated only)**.** The odd-modulus analogue (class sections at modulus q odd, e.g. $q = 3$) transports onto ratio-2 windows at non-dyadic anchors with $O(1)$ seam dust and dies modulo the off-diagonal lane write-ups (BRIDGE1-AF + GAP-N2-UNIF + Hypothesis 13 at $C = 2$). Not claimed here. [**stated**]

Source: notes/82 §2 (complete hand proofs of Q/ALT-DEAD/HSPLIT + the consistency checks §2.4), notes/80-pincer §3 (L-NOTAIL); adversarial machine verification notes/81 §2 (e186: chart exactness 44/44, AP transport 44 400/44 400, units-in-rung 6/6, fresh rung UNSAT at $p = 13$, witness 4-purity, Geneson-W Λ-scan 0 hits $t = 5..200$). Rider-free after notes/86 §1.

6 Case 2, demand I: the coupled two-seam core

The finite engine. CI(m): three blocks $(m, 8m]$, both teams block-ordered at both seams, presence bounds (balanced or constant).

Proved skeleton to transcribe: Lemma U, the anchor comparisons A1–A9, E2/C, P′, W, PAR, FG-high, Theorem H, Lemma J, the DICH case tree, Lemma D3, Lemma PH+, ASM′/COV-W′ compositions; Lemma PURE (class-c subsystem = halved double fan) and Theorem P-ARM‴; Theorem AFF$^+$ + Lemma MON. [**PROVED**] Machine layer: CI(m) UNSAT at $m = 16, 20, 24, 28, 32, 48, 64, 80$; bridge chains at six scales; law families at eight scales with blind hits. [**MACHINE-CHECKED**]

Hypothesis 20 (N6a closure)**.** The coupled core fires at every anchor: CI(m) is inconsistent for all $m \geq 16$ (balanced bounds) and all $m \geq 48$ (constant $(2, 2, 2)$ bounds). [**GAP — uniformization pool: GAP-RES (classify SAT-alive fan pairs uniformly in window length) + ThW1′-ROBUST/-TOL + DICH-F2/CASC + SPLIT + LLOP-α/β + ASM′ = (OV-$\forall$); authoritative inventory notes/77 §7.**]

Source: notes/51, 55, 56, 57, 58, 59, 66, 77; audits notes/60/61-2/76.

7 Case 2, demand II: the telescope and the demand theorem

To transcribe: the 2-adic anchor chain; T-LEDGER (disjoint 4-adic payment bookkeeping), T-FRESH (fresh mints, modulo GAP-F-schema), L-HOME/L-2PRICE/ L-SQUEEZE, J-DOWN (three-line restriction collapse), the boot-window floors J-BOOT and F-BOOT ([**PROVED**] modulo the non-gating counting gaps GAP-CMIN/FTOT), and the composition:

Theorem 21 (T-TEL″, demand side)**.** *Assume Hypothesis 20. Then every valid Case-2 pair pays, along every 2-adic anchor chain, an infinite pairwise-disjoint system of forced fresh inversion pairs — at density ≥ 1 per two octaves (and 1 per octave modulo GAP-F-schema) — each mint convertible (P3) to a displaced value in the donation ledger. No budget hypothesis is used.* ***[PROVED modulo Hypothesis 20]***

Source: notes/70 (ledger lemmas), notes/71 (boot window), notes/72 §6 (the composed statement + link-status table), notes/75 (J-DOWN, P-CAT, LEAK, NEST).

8 The corner, and the supply question

What a YES would now require. To transcribe: the dodger-corner axes (i)/(ii)/(iii) and their finite inhabitation (e179/e185 witnesses); the mod-4 lattice characterization of all realized inhabitants; Theorem AFFORD-CORNER / AFFORD-DEMAND ($\Theta(m^2)$ seam inversions and presence-scale displaced sets per anchor on the lattice family, $\Theta(N)$ cumulative) [**PROVED**]; Lemmas γ-RIGID, MINT-1, D-FLOOR/D-SAT, L-DOUBLE-DUTY [**PROVED**]; the sparse-core demand strengthening (GAP-SPARSE-CORE machine family + the AAA covering closure for $m \geq 26$) [**PROVED + MACHINE-CHECKED**]; the no-go results NG1–NG4 [**PROVED**]. By §5 the entire arithmetic corner is dead; what survives is exactly:

Hypothesis 22 (AFFORD′, the terminal supply cap)**.** No valid Case-2 pair can fund the T-TEL″ mint system — at least one displaced value per two octaves, in single-use colored values — forever. Sharpest surviving special case (all *eventually* 2^k*-periodic* shapes are closed by §5; hereditary 2-adic splitness does *not* imply aperiodicity — the minority $3\mathbb{Z}^+$ is fully periodic yet has every mod-2^k class-section bichromatic, since $\gcd(2^k, 3) = 1$; odd-periodic shapes are excluded only via Corollary 18 at constant ownership or modulo Remark 19): the cap for gap-≥ 3, 2-adically split block minorities, of which no inhabitant satisfying all corner axes is known. [**GAP — the problem's terminal open statement**]

Source: notes/62 (AFFORD′ isolation, NG4), notes/79 (tournament), notes/80 + notes/80-pincer (adjudication, pincer, AFFORD-CORNER), notes/81/82 (chart kill + SPLIT residue).

9 The conditional assembly (Theorem B)

Theorem 23 (Assembly)**.** *Assume Hypotheses 13, 20 and 22. Then no partition of* $\mathbb{Z}^+$ *into two sets has both parts 3-permutable.*

Proof shape. By Proposition 1, Case 1 or Case 2. Case 1 dies by Lemma 8 + Theorem 11, whose hypothesis (H1) is Theorem 7 at $C = 0$ plus Hypothesis 13 at $C > 0$. In Case 2, Hypothesis 20 closes the coupled core at every large anchor; Theorem 21 forces the infinite disjoint fresh-payment system; Hypothesis 22 caps supply below it. □

Dependency graph

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                 N4 dichotomy [P]
                /                \
        Case 1                    Case 2
          |                          |
   PIN [P]  DIAG-DENSE [P]   CI(m) core: skeleton [P]
          \   |                machine x8 scales [M]
           B1 [P mod H1]         |    ... Hyp N6a [GAP]
          /       \              |
  C3(p) [P]     Hyp N3-GROW    T-TEL'' demand [P mod N6a]
 (Z/D/E(p)/P     (N3-b) [GAP]    |
  L1(p), FLIP(p))                |
          |                    Hyp AFFORD' [GAP]
   B1 at C0=0 [P]              (residue: SPLIT corner;
          |                     all arithmetic shapes dead
   Lemma Q [P]                  via Lemma Q/ALT-DEAD)
          |                        |
   ALT-DEAD + HSPLIT [P]  ---->  kills the whole known
   (+ L-NOTAIL [P])              YES-corner at omega
          \                       /
           +---- Theorem B [conditional] ----+
   Theorem A = Q/ALT-DEAD/HSPLIT/L-NOTAIL, unconditional
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[P] = [**PROVED**], [M] = [**MACHINE-CHECKED**], [GAP] = open Hypothesis.

10 Machine verification and reproducibility

To transcribe from the certificates index (notes/50 §7): the strict schema executors (e113/e123 discipline), independent closure engines, independent SAT encoders and cross-validations, the two full adversarial audit cycles plus the final spot-audit (zero structural breaks), and the e186 adversarial verification of §5. Data availability mirrors Part I.

Section-to-notes map (assembly checklist)

Section	finished mathematics in	state
§2	notes/43, 46, 50 §0; Part I	transcribe
§3	notes/78 Part I + notes/33 + notes/86 §1	transcribe (prose pass done)
§4	notes/52 + notes/74 §I.4 + notes/73	transcribe
§5	notes/82 §2 + notes/81 §2 + notes/80-pincer §3	transcribe (rider-free)
§6	notes/51, 55–59, 66, 77	transcribe + Hyp 20 open
§7	notes/70, 71, 72 §6, 75	transcribe
§8	notes/62, 79, 80, 80-pincer, 81, 82	transcribe + Hyp 22 open
§9	notes/50 §5	transcribe
§10	notes/50 §7 + audits 60/60-1/61-2/76/80	transcribe

A publishable Part II exists at two truncation points: (a) *now*, as "structure + Theorem A": §§2–5 are unconditional and complete (this is the recommended cut); (b) in full, once any of Hypotheses 13–22 moves.

Data availability

As in Part I: repository with SAT certificates, schema executors, audit scripts, and a one-command reproduction kit.

References

[1] J. A. Davis, R. C. Entringer, R. L. Graham, G. J. Simmons, *On permutations containing no long arithmetic progressions*, Acta Arith. 34 (1977), 81–90.

[2] P. Erdős, R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monogr. Enseign. Math. 28 (1980).

[3] J. Geneson, *Forbidden arithmetic progressions in permutations of subsets of the integers*, 2018.

[4] T. D. LeSaulnier, S. Vijay, *On permutations avoiding arithmetic progressions*, 2010.

[5] M. Hirose, S. Saito, arXiv:2404.13510.

[6] W. Kasel, *Structural rigidity in the Erdős–Graham two-set permutation problem* (Part I), draft 2026.